Week 17: Machine memory

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

The answer keys, proofs, held hints and witness machines are in the print guide, week-17-facilitator.pdf, whose source is editable/facilitator-src/guide.json with route-note.json.

Problems and routes

Problem	K–1	2–3	4–5	route
Run a two-circle machine on rows of cards	P1			first
Rows that make exactly one of two machines say yes		P1		first
Which rows does a three-circle machine accept?			P1	first
Every four-card row that finishes at yes	P2			first
Rows that give two machines different answers	P3			first
Has a red card appeared?		P2		first
Is the last card blue?	P4	P3		K-1 return; 2-3 first
A machine for no red cards	P5			return
Can the same added cards separate two rows?	P6	P4	P2	K-1 return; 2-3 first; 4-5 first
Two rows on the same circle			P3	return
The fewest circles for groups of three		P5		return
The fewest circles for groups of four			P4	return
The fewest circles for rows ending red, blue		P6	P5	return
The fewest circles for any group size			P6	return

First route. K–1 P1–P3; Grades 2–3 P1–P4; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P6; Grades 2–3 P5–P6; Grades 4–5 P3–P6.

Key, one entry per problem

Run a two-circle machine on rows of cards

K–1 P1 · first route

Idea. The machine says yes exactly when the number of red cards is even.

Rows that make exactly one of two machines say yes

Grades 2–3 P1 · first route

Idea. A four-card row has an even number of reds or ends in red, but not both, in exactly 8 of 16 rows.

Which rows does a three-circle machine accept?

Grades 4–5 P1 · first route

Idea. The three-circle cycle accepts exactly the rows whose number of red cards is a multiple of three.

Every four-card row that finishes at yes

K–1 P2 · first route

Idea. This machine says yes exactly when the last card is red: 8 of the 16 four-card rows.

Rows that give two machines different answers

K–1 P3 · first route

Idea. Two different machines can be told apart by some row.

Has a red card appeared?

Grades 2–3 P2 · first route

Idea. Two circles suffice and are needed: a permanent memory of whether red has appeared.

Is the last card blue?

K–1 P4 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

Idea. Two circles: each card resets the answer, so only the last card matters.

A machine for no red cards

K–1 P5 · return route

Idea. Two circles: once a red card appears the answer is no forever.

Can the same added cards separate two rows?

K–1 P6 · Grades 2–3 P4 · Grades 4–5 P2 · K-1 return; 2-3 first; 4-5 first route · wording differs by band

Idea. Two rows that leave a machine on the same circle can never be separated by a shared ending; rows separated in this way need different circles.

Two rows on the same circle

Grades 4–5 P3 · return route

Idea. Rows that end on the same circle get the same answer after any shared ending.

The fewest circles for groups of three

Grades 2–3 P5 · return route

Idea. Three circles: the rows with 0, 1 and 2 reds can be separated two at a time.

The fewest circles for groups of four

Grades 4–5 P4 · return route

Idea. Four circles: the rows with 0, 1, 2 and 3 reds can be separated two at a time.

The fewest circles for rows ending red, blue

Grades 2–3 P6 · Grades 4–5 P5 · return route · wording differs by band

Idea. Three circles: the empty row, R and RB can be separated two at a time.

The fewest circles for any group size

Grades 4–5 P6 · return route

Idea. For groups of m, m circles suffice and are needed (the easy half of Myhill-Nerode).